Fiscal Year:	FY 2021	**Task Last Updated:**	FY 02/02/2022
PI Name:	Vekilov, Peter G. Ph.D.		
Project Title:	Solution Convection and the Nucleation Precursors in Protein Crystallization		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77204-4004	**Congressional District:**	18
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
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No. of PhD Candidates:	5	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA MSFC
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Conrad, Jacinta Ph.D. (University of Houston) Maes, Dominique Ph.D. (Vrije Universiteit Brussel)		
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Task Description:	Protein crystallization experiments in microgravity have often generated unexpectedly low or high numbers of crystals; both outcomes may doom experiments designed to obtain a small number of well-separated crystals for x-ray structure studies. This unpredictability has been associated with the suppression of buoyancy-driven convection, as supported by data indicating that protein crystal nucleation rates reach a maximum at an optimal convection velocity. To elucidate the effects of solution convection on crystal nucleation--a crucial issue in several fields of science and engineering--we propose to test two mechanisms: 1. Convection enhances or suppresses the formation of the dense liquid clusters, within which crystal nuclei form. 2. Convection enhances or suppresses the formation of crystal nuclei within the clusters. Tests of mechanism 1. require characterization of the dense liquid clusters at different rates of convection, including its complete absence, only possible in microgravity. We propose to use the Light Microscopy Module (LMM) aboard the International Space Station (ISS) and two variants of the Advanced Colloid Experiment (ACE): ACE-M and ACE-C. To prepare for microgravity experiments, we will design microscopy techniques that can then be implemented on the LMM. We will extend a newly-developed method, differential dynamic microscopy, to quantify the dynamics of the dense liquid clusters. Its variation with solution composition and the rate of convection will be used to identify the cluster formation mechanisms. Towards tests of mechanism 2, we will determine the supersaturation dependencies of the crystal nucleation rates in both quiescent and flowing solutions. For this, we will implement novel microfluidics devices made of SU-8 and glass, which allow long-duration experiments under steady conditions. Finally, we will use the insight on convection effects on nucleation to establish procedures leading to higher perfection of protein crystals and test them with proteins of interest to biomedical sciences.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Solution crystallization underlies a broad range of industrial, laboratory, and physiological processes. Single solution-grown crystals of inorganic salts or mixed organic-inorganic materials are used in non-linear optics elements and for other electronic and optical-electronic applications; chemical products and production intermediates are precipitated as crystals in thousands-of-tons amounts. Another area which relies on solution-grown crystals is pharmacy: the slow crystal dissolution rate is used to achieve sustained release of medications: small-molecules organic, or proteins such as insulin, interferon-alpha, or the human growth hormone. If the administered dose consists of a few equidimensional crystallites, steady medication release rates can be maintained for longer periods than for doses comprised of many smaller crystallites. The formation of protein crystals and crystal-like ordered aggregates underlies several human pathological conditions. An example is the crystallization of hemoglobin C and the polymerization of hemoglobin S that cause, respectively, the CC and sickle cell diseases. The formation of crystals in the eye lens underlies the pathology of several forms of cataract. A unique example of benign protein crystallization in humans and other mammals is the formation of rhombohedral crystals of insulin in the islets of Langerhans in the pancreas. Traditionally, protein crystals have been used for the determination of the atomic structure of protein molecules by x-ray crystallography; this method contributes ~87% of all protein structures solved, with the majority of the other determinations carried out by Nuclear Magnetic Resonance (NMR) spectroscopy. Due to its crucial place at the start of the crystallization process, the nucleation of crystals determines many properties of the emerging crystalline phase. It is obvious that the nucleation selects the polymorphic form and if a different polymorph is desired, conditions at which its nucleation is faster than that of the other possible polymorphs should be sought. If nucleation is fast, many crystals form nearly simultaneously. Their growth depletes the medium of solute and may lead to cessation of nucleation at the later stages of crystallization. Thus, the majority of crystals grow to approximately identical sizes. In contrast, if nucleation is slow and fewer crystals nucleate at a time, the supersaturation in the solution drops slowly, the nucleation of new crystals continues and a population of crystals of various sizes forms. Ultimately, if nucleation is hindered everywhere in the growth container but at a few selected spots, crystals only nucleate at these spots and grow large before the solution is depleted of nutrient. Hence, control of nucleation is a means to control size, size distribution, polymorphism and other properties of the crystals.
Task Progress:	No reporting received from investigator. See Bibliography section below for publications from 2017-present. The Cumulative Bibliography will display all publications resulting from this grant.
Bibliography Type:	Description: (Last Updated: 12/05/2024)
Articles in Other Journals or Periodicals	Byington MC, Safari MS, Lubchenko V, McCabe JW, Angle LA, Hawke DH, Bark SJ, Conrad JC, Vekilov PG. "Weakly-bound dimers that underlie the crystal nucleation precursors in lysozyme solutions." bioRxiv preprint server. Posted March 8, 2018. This article is a preprint and has not been certified by peer review. https://doi.org/10.1101/275222 [Ed. note February 2022: remains in preprint server] , Mar-2018
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Articles in Peer-reviewed Journals	Olafson KN, Ketchum MA, Rimer JD, Vekilov PG. "Molecular mechanisms of hematin crystallization from organic solvent." Crystal Growth & Design. 2015 Nov 4;15(11):5535-42. https://doi.org/10.1021/acs.cgd.5b01157 , Nov-2015
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Articles in Peer-reviewed Journals	Yamazaki T, Kimura Y, Vekilov PG, Furukawa E, Shirai M, Matsumoto H, Van Driessche AE, Tsukamoto K. "Two types of amorphous protein particles facilitate crystal nucleation." Proc Natl Acad Sci U S A. 2017 Feb 28;114(9):2154-9. https://doi.org/10.1073/pnas.1606948114 ; PMID: 28193873; PMCID: PMC5338535 , Feb-2017
Articles in Peer-reviewed Journals	Olafson KN, Nguyen TQ, Vekilov PG, Rimer JD. "Deconstructing quinoline-class antimalarials to identify fundamental physicochemical properties of beta-hematin crystal growth inhibitors." Chemistry. 2017 Oct 4;23(55):13638-47. https://doi.org/10.1002/chem.201702251 ; PMID: 28833627 , Oct-2017
Articles in Peer-reviewed Journals	Olafson KN, Rimer JD, Vekilov PG. "Early onset of kinetic roughening due to a finite step width in hematin crystallization." Phys Rev Lett. 2017 Nov 10;119(19):198101. https://doi.org/10.1103/PhysRevLett.119.198101 ; PMID: 29219496 , Nov-2017
Articles in Peer-reviewed Journals	Safari MS, Byington MC, Conrad JC, Vekilov PG. "Polymorphism of lysozyme condensates." J Phys Chem B. 2017 Oct 5;121(39):9091-101. https://doi.org/10.1021/acs.jpcb.7b05425 ; PMID: 28881129 , Oct-2017
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Articles in Peer-reviewed Journals	Ma W, Lutsko JF, Rimer JD, Vekilov PG. "Antagonistic cooperativity between crystal growth modifiers." Nature. 2020 Jan;577(7791):497-501. https://doi.org/10.1038/s41586-019-1918-4 ; PMID: 31942074 , Jan-2020
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Articles in Peer-reviewed Journals	Yang DS, Saeedi A, Davtyan A, Fathi M, Sherman MB, Safari MS, Klindziuk A, Barton MC, Varadarajan N, Kolomeisky AB, Vekilov PG. "Mesoscopic protein-rich clusters host the nucleation of mutant p53 amyloid fibrils." Proc Natl Acad Sci U S A. 2021 Mar 9;118(10):e2015618118. https://doi.org/10.1073/pnas.2015618118 ; PMID: 33653952; PMCID: PMC7958401 , Mar-2021
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Dissertations and Theses	Safari MS.. "Polymorphism of Protein Condensates." Dissertation, University of Houston, May 2018. , May-2018